

Aditya Verma

B.Tech in Computer Science and Engineering (Data Science)

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Summary

Computer Science undergraduate with strong coding skills (CodeChef 3★, Global Rank 34) and practical experience in full-stack development and machine learning. Proficient in designing and deploying scalable web applications using React, Node.js, and MongoDB, with additional expertise in building ML models for real-time applications.

Education

Techno Main Salt Lake — B.Tech CSE (Data Science)

| GPA: 7.48/10 | Expected: May 2026

Technical Skills

- Languages: C++, Python, Java, C
- Libraries/Frameworks: TensorFlow, Pandas, NumPy, Matplotlib, Scikit-learn
- Web Technologies: HTML, CSS, JavaScript, React.js, Node.js, Express.js, Tailwind CSS
- Database: MongoDB, MySQL, PostgreSQL
- Tools & Platforms: Git, GitHub, VS Code, Postman, JWT, Tableau
- Core Concepts: DSA, OOP, OS, DBMS, APIs, CN, System Design

Projects

Sneaker Hub |

[Link](#) | React, Node.js, Express, MongoDB, JWT

- Engineered secure user authentication flows (JWT, OAuth2.0) within the e-commerce application, contributing over 6000 lines of code and enabling seamless login experiences for all users.
- **Developed and deployed a scalable REST API using Node.js and MongoDB, cutting API latency by 20%** through defined database queries and caching strategies, resulting in faster response times.
- **Designed and automated unit & functional test suites**, significantly improving **code coverage, reliability, and regression detection**.

Emotion Recognition System |

[Link](#) | Python, Tensorflow, OpenCV

- Engineered a real-time facial emotion recognition system leveraging a custom **CNN** architecture trained on the **FER2013 dataset**.
- Optimized model performance via hyperparameter tuning (learning rate schedules, dropout, batch normalization) and data augmentation, achieving **64%** validation accuracy.
- Deployed the solution with OpenCV integration for live webcam inference, enabling interactive, real-time emotion classification.

VAANI(Voice and Audio Analysis Natural Interface)|

[Link](#) | Python, Tensorflow, Conv1D, BiLSTM

- Developed a real-time speech emotion and speaker recognition system using **MFCC**-based audio features and deep learning models.
- Designed and trained **Conv1D + BiLSTM + Attention networks** to classify human emotions and identify speakers from live microphone input.
- Implemented real-time audio capture, preprocessing, and inference pipeline with noise-robust augmentation and streamlined training strategies.
- Achieved high accuracy using multi-dataset training (RAVDESS, TESS, CREMA-D, SAVEE) and advanced learning rate scheduling techniques.

Programming Achievements

- ❖ Rated 3 ★ on CodeChef with Rating 1616. [CC Profile](#)
- ❖ Rated Pupil on Codeforces. [CC Profile](#)
- ❖ Secured Global Rank 34 in Global Starters 158 on Codechef among 10000+ participants
- ❖ Solved 500+ questions on different coding platforms and online judges
- ❖ Advanced through the first round of Flipkart Grid 2024; participated in Smart India Hackathon 2024

Social Engagements

Club Membership: Active member of Techno Main Salt Lake's Sport Activity Club.

Sports-Engagements: Cultivated discipline and teamwork through consistent participation in team sports, attending an average of 3 sessions per week over 2 years, enhancing collaborative skills.

Reading: Synthesized insights from books across diverse genres, including biographies and technology, to formulate three novel strategies presented during team brainstorming sessions contributing to increased innovation.